

● HARD

● GEOMETRY AND TRIGONOMETRY

# Geometry and Trigonometry

# 13

4 questions · ~6 min

Q1

GEOMETRY AND TRIGONOMETRY

$$(x-6)^2 + (y+5)^2 = 16$$

In the  $xy$ -plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates  $(10, -5)$ . If  $\overline{PQ}$  is a diameter of the circle, what are the coordinates of point Q ?

- A.  $(2, -5)$
- B.  $(6, -1)$
- C.  $(6, -5)$
- D.  $(6, -9)$

**Q2**

GEOMETRY AND TRIGONOMETRY

An isosceles right triangle has a hypotenuse of length 58 inches. What is the perimeter, in inches, of this triangle?

- A.  $29\sqrt{2}$
- B.  $58\sqrt{2}$
- C.  $58 + 58\sqrt{2}$
- D.  $58 + 116\sqrt{2}$

**Q3**

GEOMETRY AND TRIGONOMETRY

The equation  $x^2 + y - 1^2 = 49$  represents circle A. Circle B is obtained by shifting circle A down 2 units in the  $xy$ -plane. Which of the following equations represents circle B?

- A.  $x - 2^2 + y - 1^2 = 49$
- B.  $x^2 + y - 3^2 = 49$
- C.  $x + 2^2 + y - 1^2 = 49$
- D.  $x^2 + y + 1^2 = 49$

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

Point  $O$  is the center of a circle. The measure of arc  $RS$  on this circle is  $100^\circ$ . What is the measure, in degrees, of its associated angle  $ROS$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.